MINING DEALER BEST PRACTICE

Electronic Engine Simulation Board

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

Training room capability is one of the very important keys to perform technical training in an efficient way. Typically TCDP requires classroom and on-the-job training for each competency level. The classroom trainings give theoretical information for electrical system of the machine and many trainees cannot imagine the functionality of the system that is written in the documentation. Since the availability of finding a real engine during classroom training is nearly impossible, therefore the combination of both classroom and on-the-job training is very crucial. The Cat Electronic Engine Simulation Board has been designed to find an optimum solution for cost savings and to combine classroom training with on-the-job training to deliver the efficient training sessions for trainees.

2.0 Best Practice Description

The CAT Electronic Engine Simulation board is designed to get same electrical system functionality of C9 engine. The board consists of:

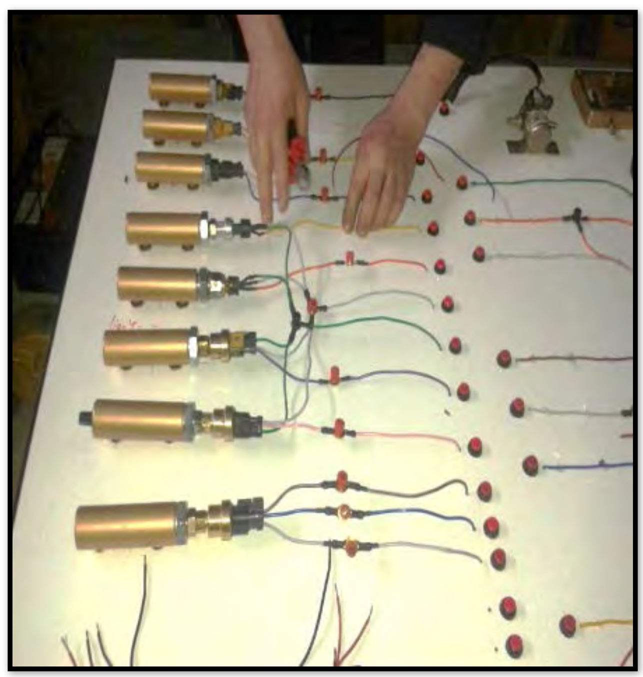
Injectors
ECM
Sensors, switches and relays
Gear motor to trigger the gear where speed sensor equipped
Excavator monitoring control module
Product link control module
The connection cable net with electrical parts to run the system
Test points according to electrical schematic of the engine
ET connection port
Visual camera
Etc…

The board function is like an engine. When the start key is turned on, all components are energized and run like a real engine electrical circuit. It allows trainees to check functionality, test the circuit, measure the voltage, current and resistance, visually understand the electrical flow priorities and detect the problem. The board also provides failure simulation for trainees to be aware of usual problems and to check the diagnostic codes when the system generates one.

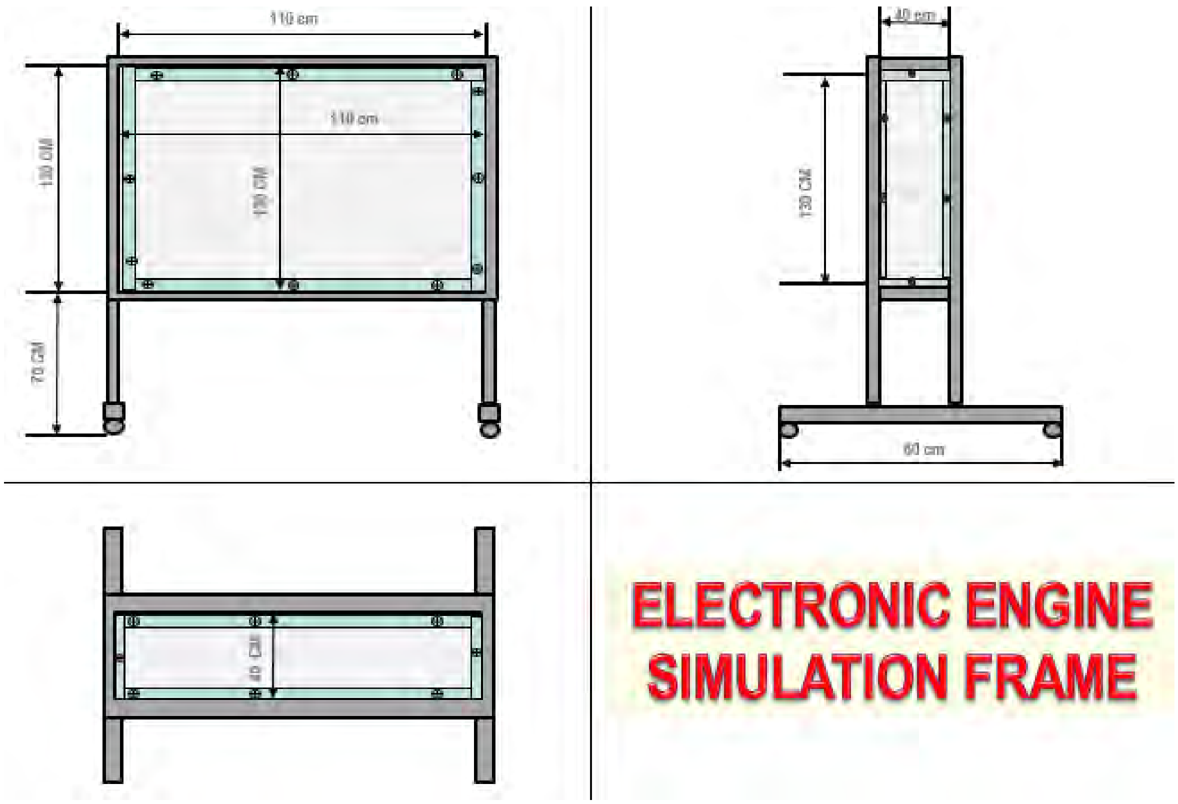
3.0 Implementation Steps

Step-1 is to build holder of sensors, injectors and HEUI pump.

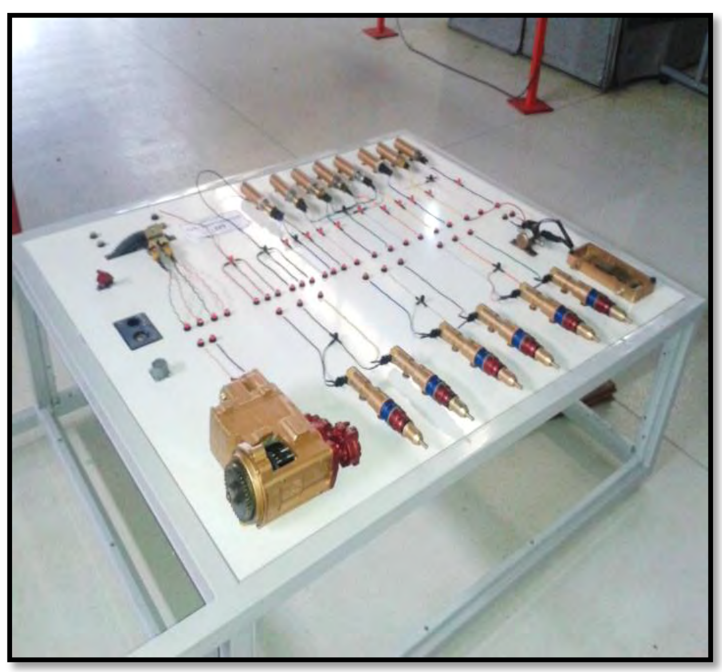
Step-2 is to allign all componets to board and connect cables according to electrical scematic



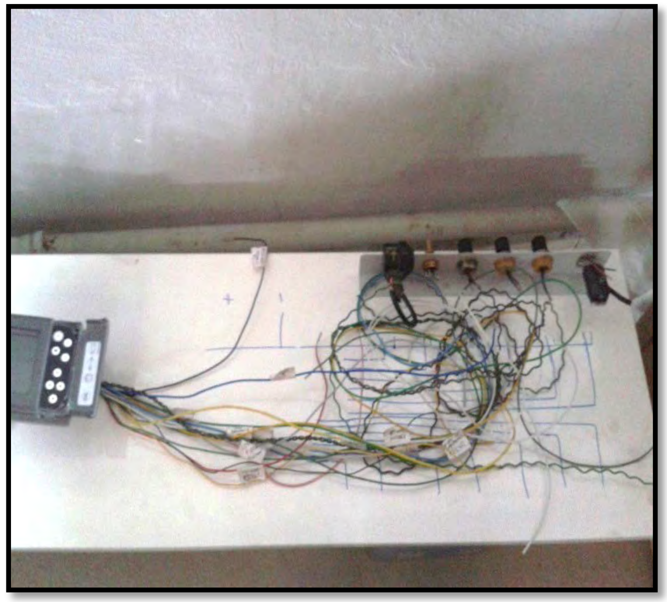
Drawing and dimension of the main body is following;



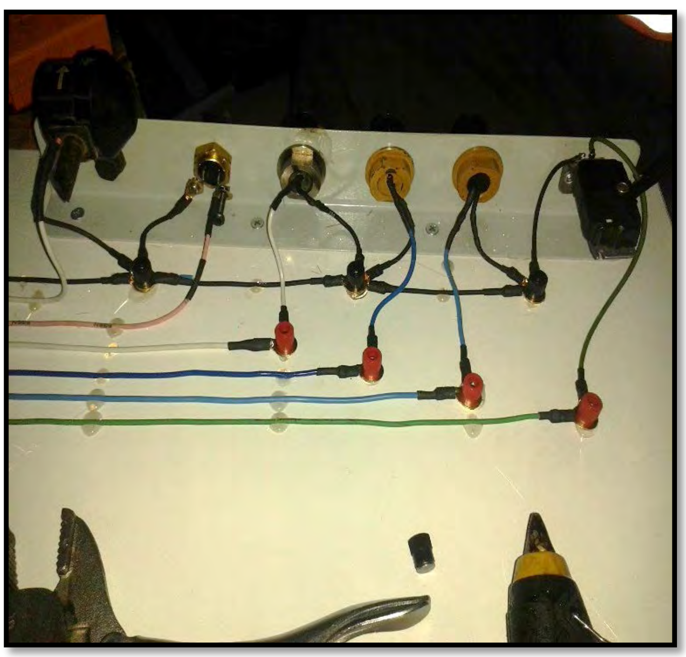
The next step is to put the electrical component board and main body together

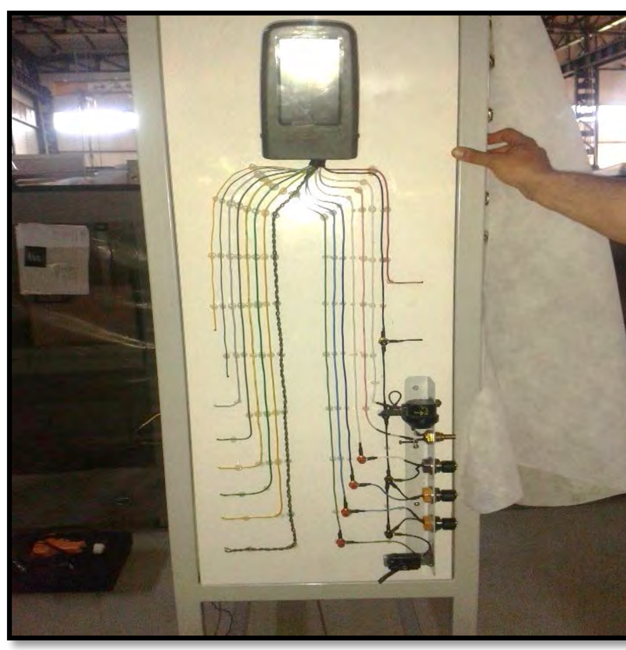


Hydraulic excavator monitoring system has been designed with level sensors, action alarm and rear camera. The following picture is how to build excavator monitoring simulation board.

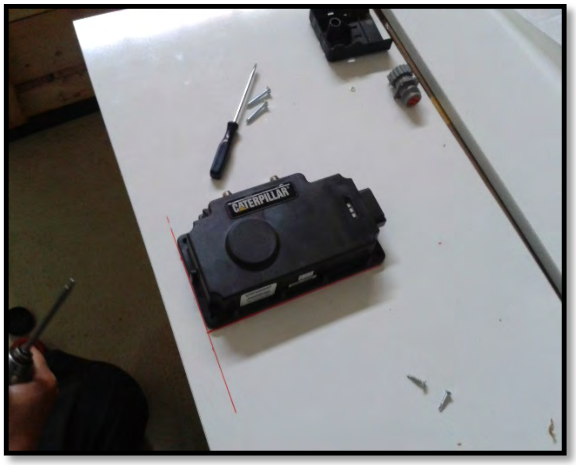


All monitor, sensors and warning alert modules were mounted to the plate. The wiring connection was designed according to electrical schematic.
Measurement jacks have been located in the circuit to test the individual lines

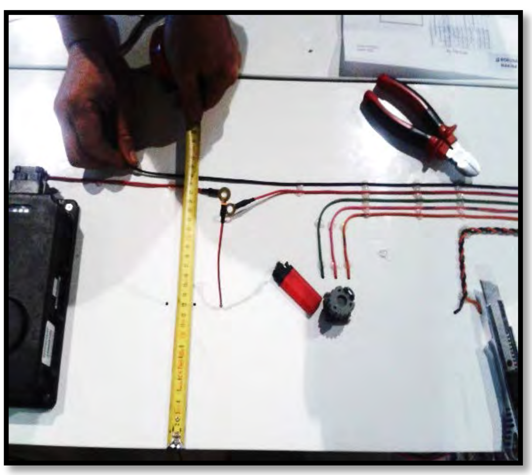


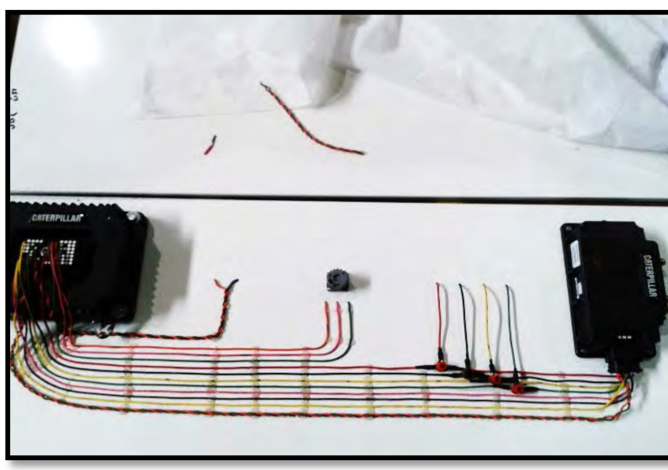


Product Link system PL300 ECM module was mounted to the left side of board and PL100 ECM to the right side of the main board

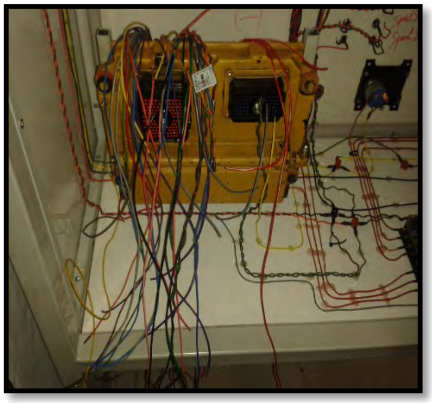


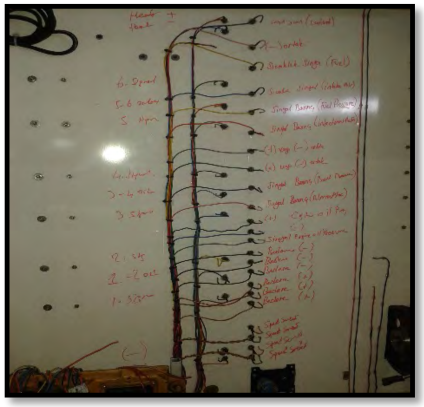
The wiring harness was designed according to Product Link electrical schematic

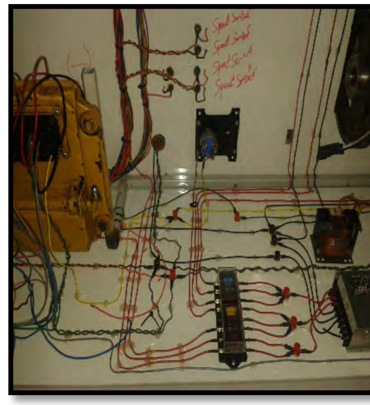




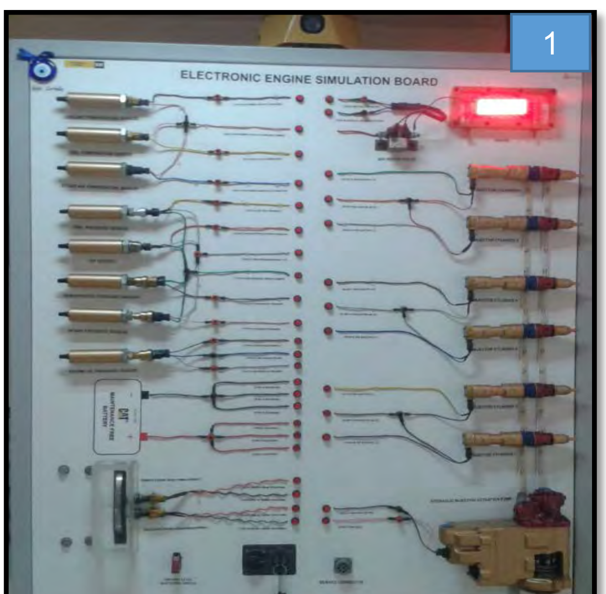
The engine ECM, power unit, gear motor assembly and wiring harness were located into rear side of the main board

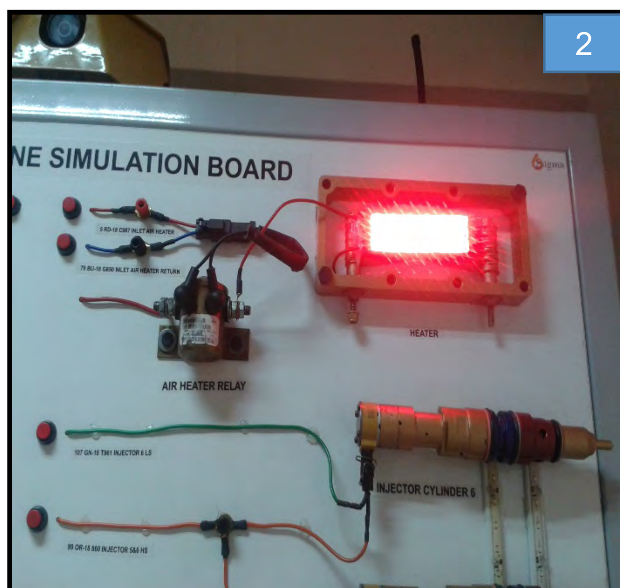






The LED system has been designed to illustrate functionality of the heater lines, fuel line of injectors, and oil line between HEUI pump and injectors.





How does it work?

- Start operation when the key turned on
- Gear motor starts to run
- Timing sensor equipped on the output shaft gear generate signal to the engine ECM
- When the RPM on gear motor output shaft reaches 170rpm, the engine ECM sends energy to the injectors
- Module generates sound similar a diesel engine

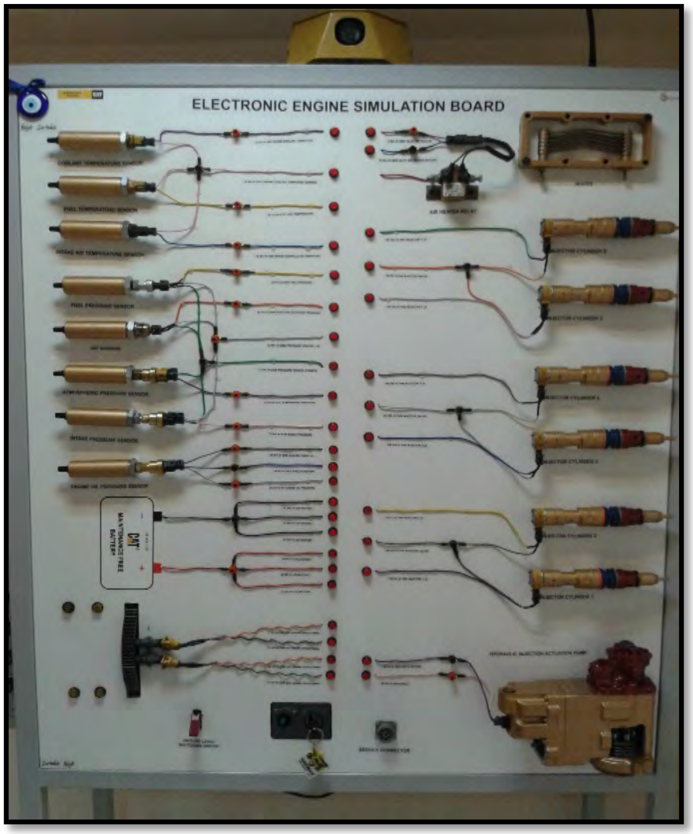
Rear Section

Front Section

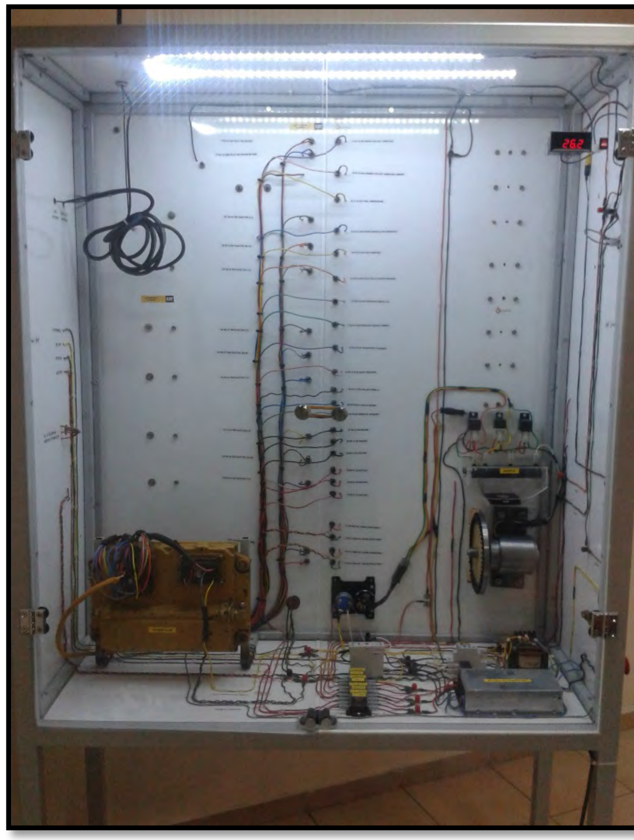
There are two different colors led the line to injectors.

- Blue led flash is the line of HEUI oil
- Red led flash is the line of fuel

When the key is switched on, the LED flash illustrates flow of oil and fuel to injectors in an alignment.

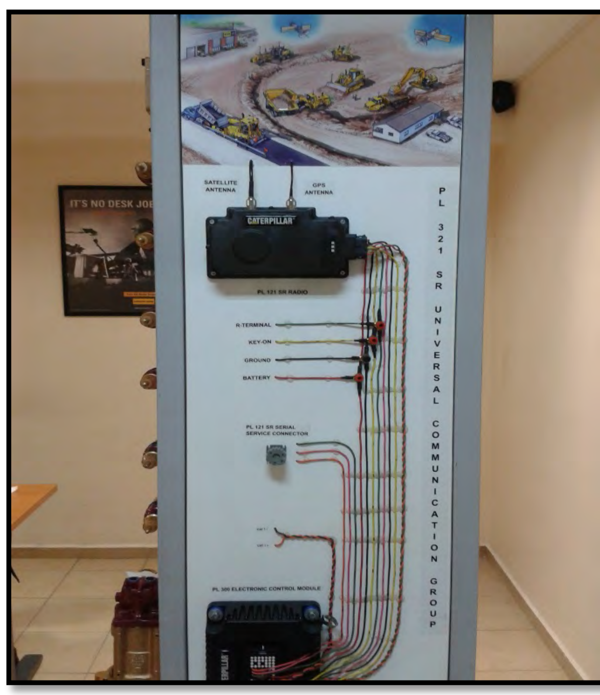


Main front board



Rear Section

Excavator Monitoring Simulator



Product Link Simulator

4.0 Benefits


IT'S NO DESK JOB.
ELECTRONIC ENGINE SIMULATION BOARD


IT'S NO DESK JOB.


TCDP ELECTRONIC ENGINE
SIMULATION BOARD

The following trainings can be given to technicians and engineers:

1) Basic Electric and Electronic training
   a) Reading electrical schematic
   b) Electric parts and remits
   c) Using multimeter
   d) Serial and parallel circuit
   e) Making diagnostic codes and learn strategy in electronic systems
2) Electronic Technician Program Training
   a) How to use electronic technician program
   b) Loading flash program
   c) Adjust configuration parameter on ET program
   d) Find and remove active codes and event codes
   e) View status parameter
   f) Using override menu
   g) Use Data log and real time graphic
3) Electronic Engine Training
   a) System operation of electronic engine
   b) Demonstrate input and output devices
   c) Evaluation of engine performance specifications
4) CAT hydraulic excavator monitoring system training
   a) Demonstrate monitor menu
   b) Demonstrate active code and logged code
   c) Demonstrate calibration solenoid and sensors
   d) Demonstrate configuration parameters
5) Product link training
   a) Demonstrate how to register
   b) Assembled and control PL ECM
   c) How to use Product link parameters
6) HEUI fuel system training
   a) Demonstrate part of fuel system
   b) Testing HEUI pump
   c) Demonstrate injector solenoid
   d) Demonstrate cylinder cutout test
7) Electrical and electronic failure analyze training
   a) Demonstrate active codes and solve problem
   b) Demonstrate logged codes
   c) Demonstrate active event codes
   d) Demonstrate logged event codes
8) TCDP basic electrical and monitoring systems on the job training

5.0 Resources Required

Total cost of investments:

Item		Cost (USD)
Material		$ 3,800
Cat Parts		$ 13,000
Labor	Design	$ 2,500
	Assembly	$ 1,300
Total		$ 20,600

6.0 Supporting Attachments / References

- English EESB TOOL OPERATION MANUEL BMGS001.pdf

7.0 Related Best Practices

None

8.0 Acknowledgements

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